Samantha Ellis

NSF Graduate Research Fellow
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Education

- **California State University, Northridge** — *Northridge, USA*
  Master of Science, Chemistry (GPA: 3.92) — Expected Graduation Spring 2027
  Bachelor of Science, Chemistry (GPA: 3.77) — 2024
- **University of California, Irvine** — *Irvine, USA*
  Completed coursework toward Bachelor of Science, Chemistry — 2019–2020
- **Fullerton College** — *Fullerton, USA*
  Associate of Arts, Chemistry — 2019

Research Experience

- **Research Student Internship, Oak Ridge National Laboratory** — *May 2026 – Present*
  Advisors: Eva Zarkadoula and Jingsong Huang
  - Use DFT reaction pathways, CI-NEB activation barriers, and ICOHP bonding analysis to evaluate and interpret MLIP-predicted etching chemistry in MoS_2 and WS_2, connecting atomic-scale mechanisms to experimental etching differences.
  - Fine-tune and validate material-specific MACE-MH-1 potentials against DFT to produce an accurate MLIP to model reactive 2D etching chemistry.
  - Develop active-learning workflows combining reactive molecular dynamics, targeted DFT labeling, and iterative fine-tuning to expand coverage of evolving pore environments.
  - First-author manuscript in preparation.
- **Graduate Research Assistant, CSUN** | Advisor: Maosheng Miao — *January 2025 – Present*
  Realizing Pressure-Induced Cs 5p Core Reactivity in Cs Polyoxides
  - Develop computational guidance for experimental target selection and spectroscopic identification of core-reactive Cs polyoxide phases under high pressure as the basis of my M.S. thesis in Chemistry.
  - Identified spectroscopic signatures of pressure-induced Cs 5p core reactivity in CsO using XANES calculations to guide experimental diamond anvil cell studies.
  - Built ternary Cs polyoxide convex hulls at 150 and 250 GPa to screen 35–45 candidate structures per hull and narrow experimentally viable reaction chemistries.
  - First author manuscript in preparation.

  High-Pressure Redox Chemistry of Iron in Deep Earth Systems
  - Mapped pressure-dependent phase stability across Fe–p-block compositions at 0, 150, and 300 GPa through CALYPSO structure searches and 25 HSE-based convex hulls.
  - Quantified pressure-induced reversal of Fe redox behavior using HSE-based Bader charge analysis of 60 structures.
  - Co-authored publication.

  Molecular Hydrogen Insulator-to-Metal Transition under High Pressure
  - Characterized pressure-dependent band-gap closure and metallization in candidate molecular hydrogen phases using PBE and HSE band-structure calculations.
  - Evaluated whether metallization can precede atomization through wavefunction-based analysis of the electronic character of compressed molecular hydrogen.
  - Co-authored publication.
- **Undergraduate Research Assistant, CSUN** | Advisor: Maosheng Miao — *January 2024 – January 2025*
  - Conducted quantum-chemistry simulations on Linux-based HPC clusters using VASP and particle-swarm optimization to study electronic structure in solid-state systems.
- **Undergraduate Research Assistant, CSUN** | Advisor: Taeboem Oh — *January 2023 – January 2024*
  - Supported multistep synthesis and NMR characterization of small organic molecules, including reaction setup and purification workflows.
- **Undergraduate Research Assistant, UCI** | Advisor: Sergey Nizkorodov — *December 2019 – December 2020*

- Optimized experimental protocols and novel SOA sampling devices to improve collection efficiency and data reliability.

Publications

- **Prediction of Electron Localization Functions from Superposed Atomic Densities for Accelerated Superhydride Discovery** Ellis, A.; **Ellis, S.**; Miao, M. *J. Mater. Inf.* **2026**, *6*, 45. DOI: 10.20517/jmi.2026.44.
- **Emerging Electride Behavior and Metallization in Molecular Hydrogen under High Pressure** Ellis, A.; **Ellis, S.**; Pandit, A.; Miao, M. *J. Phys. Chem. Lett.* **2026**. DOI: 10.1021/acs.jpclett.6c01415.
- **Pressure-Induced Redox Reversal of Iron and the Distribution of Elements in Deep Earth** Wang, X.; Feng, X.; Li, J.; Lv, Y.; Ellis, A.; **Scott, S.**; Pandit, A.; Khodagholian, D.; Hemley, R. J.; Jackson, M. G.; Spera, F. J.; Redfern, S. A. T.; Miao, M. *Proc. Natl. Acad. Sci. U. S. A.* **2025**, *122*(46), e2414911122. DOI: 10.1073/pnas.2414911122.

Manuscripts in Progress

- **Cesium Core Reactivity in Cs-Ba-O Ternary Oxides Under Pressure** **Ellis, S.**; Ellis, A.; Wyke, L.; Miao, M. Manuscript in preparation.
- **Abnormal Behavior of Cs Polyoxides under High Pressure** Sun, Y.; **Ellis, S.**; Khodagholian, D.; Ellis, A.; Miller, P.; Ji, C.; Gou, H.; Dronskowski, R.; Miao, M. Manuscript submitted.

Selected Presentations

Oral Presentations:

- **S. Ellis**: *Pressure-induced Redox Reversal of Iron and the Distribution of Elements in Deep Earth*. 2026 CSUN Research and Creative Works Symposium (CSUNposium), Northridge, CA, April 2026.
- **S. Scott**: *Pressure-induced Redox Reversal of Iron and the Distribution of Elements in Deep Earth*. 2026 American Physical Society Global Summit (APS), Denver, CO, March 2026.
- **S. Scott**: *Pressure-induced Redox Reversal of Iron and the Distribution of Elements in Deep Earth*. 2025 CSUN Sigma Xi Research Symposium, Northridge, CA, December 2025.

Poster Presentations:

- **S. Ellis**, M. Lee, J. Huang, and E. Zarkadoula: *Modeling Gas-Assisted Etching of MoS_2 and WS_2 with Machine-Learned Interatomic Potentials*. 2026 Oak Ridge Institute for Science and Education (ORISE) Symposium, Oak Ridge, TN, July 2026.
- **S. Ellis**, Y. Sun, A. Ellis, X. Chen, M. Miao: *An Effective Machine Learning Frame for Materials Discovery Structured by a Chemical Concept*. 2025 NeurIPS AI4Mat Workshop, San Diego, CA, December 2025.
- **S. Scott**, A. Ellis, A. Pandit, and M. Miao: *Insulator to Metal Transition of Molecular Hydrogen Under Pressure*. 2025 CSUN Research and Creative Works Symposium (CSUNposium), Northridge, CA, April 2025.
- **S. Scott**, A. Ellis, A. Pandit, and M. Miao: *Insulator to Metal Transition of Molecular Hydrogen Under Pressure*. 2025 American Physical Society Global Summit (APS), Anaheim, CA, March 2025.

Co-authored Presentations:

- M. Lee, **S. Ellis**, E. Zarkadoula, and J. Huang: *Mechanistic Insights into Fluorine-Assisted Etching of Transition Metal Dichalcogenides*. 2026 Oak Ridge Institute for Science and Education (ORISE) Symposium, Oak Ridge, TN, July 2026.
- A. Ellis, **S. Scott**, and M. Miao: *Seitz-Invariant Prediction of Electron Localization Functions from Superposed Atomic Densities via a 3D-Convolutional Network*. 2026 American Physical Society Global Summit (APS), Denver, CO, March 2026.
- A. Pandit, A. Ellis, **S. Scott**, and M. Miao: *Interactions that Govern the Structure of α-Manganese*. 2025 American Physical Society Global Summit (APS), Anaheim, CA, March 2025.

Teaching Experience

- **Teaching Assistant - General Chemistry and Organic Chemistry Labs, CSUN** *January 2025 – May 2026*
  - Taught two General Chemistry laboratory sections for 2 semesters, guiding 24 students per section in solution preparation, titration, electrochemistry, and quantitative analysis.
  - Taught two Organic Chemistry laboratory sections for 1 semester, guiding 18 students per section in synthesis, purification, spectroscopic analysis, and mechanistic interpretation.
  - Incorporated scientific literature and research-oriented thinking into laboratory instruction through extracurricular assignments.

Mentoring and Outreach

- **Journal Coordinator - CSUN ACS Chemistry and Biochemistry Club** *August 2025 – May 2026*
  - Co-founded and coordinated CSUN's monthly Journal Club spanning chemistry and biochemistry, including article selection, presenter scheduling, and event logistics.
  - Supported early undergraduate mentorship by pairing students with graduate facilitators and connecting interested participants with research labs and opportunities.
- **Graduate Peer Mentor - Office of Undergraduate Research, CSUN** *June 2025 – May 2026*
  - Lead three weekly student-facing workshops reaching 30–45 students on research onboarding, graduate applications, conference preparation, and professional development.
  - Provided weekly one-on-one advising on joining research labs, drafting SOPs and personal statements, preparing abstracts and posters, and developing CVs and resumes.
- **Vice President - Alpha Chi Sigma Chemistry Fraternity, UC Irvine** *May 2020 – May 2021*
  - Organized research networking opportunities for members, including invited talks from UC Irvine researchers and lab tours across chemistry-related research groups.
  - Coordinated professional development workshops on resume and CV preparation to support undergraduate career and research readiness.
  - Led member recruitment and engagement efforts.

Industry Experience

- **Research and Development Lab Technician - New Look Cosmetics, Chatsworth** *May 2022 – January 2024*
  - Managed 3–5 concurrent projects and collaborated with engineers to scale validated formulations from laboratory to pilot production.
  - Designed, optimized, and troubleshot formulation experiments with chemists using quantitative and analytical problem-solving methods.
  - Maintained electronic formulation and project-management records to support reproducibility and data integrity.

Honors and Awards

- **National Science Foundation Graduate Research Fellowship (NSF GRFP)** *2026*
  National Science Foundation
- **Leslie and Terry Cutler Scholarship** *2026*
  College of Science and Mathematics, CSUN
- **Dr. Frank Harris Graduate Scholarship Endowment in Chemistry and Biochemistry** *2026*
  Department of Chemistry and Biochemistry, CSUN
- **Outstanding Achievement Award** *2026*
  Department of Chemistry and Biochemistry, CSUN
- **Outstanding Leadership Award** *2026*
  Department of Chemistry and Biochemistry, CSUN
- **Second Place, Sigma Xi Research Symposium** *2025*
  Sigma Xi Research Society, CSUN
- **Dr. Lan K. Wong Undergraduate Research Award - $1000** *2022*
  Department of Chemistry and Biochemistry, CSUN
- **ACS Undergraduate Award in Organic Chemistry** *2022*
  Department of Chemistry and Biochemistry, CSUN
- **Michael E. Gebel Award** *2020*
  Department of Chemistry, UCI